

SHUCHANG LIU

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RESEARCH EXPERIENCE

Massachusetts Institute of Technology

Cambridge, USA

Postdoctoral researcher

Jun. 2024 – present

- A physically-constrained downscaling approach based on dynamical downscaling and machine learning (funded by Swiss National Science Foundation Postdoc Mobility fellowship, CHF 128k)
- Constraining precipitation uncertainty under climate change using machine learning (MIT Climate Grand Challenges Initiative)
- Generalized convection parameterization: adapting current-climate trained models for warmer climates (Affiliate scientist of [Multiscale Machine Learning In Coupled Earth System Modeling](#))

ETH Zürich

Zürich, Switzerland

Postdoctoral researcher

Oct. 2023 – May. 2024

- How to deal with double-ITCZ when downscaling in the tropics

PhD candidate

Oct. 2020 – Sep. 2023

- The potential of high-resolution climate models in the tropics in constraining climate-change uncertainties

Tsinghua University

Beijing, China

Student Researcher

Sep. 2017 - June 2020

- The impact of climate mitigation pathways on the heatwave and air pollution-related human health in China

EDUCATION

ETH Zürich

Zürich, Switzerland

PhD Atmospheric and Climate Science

Sep. 2023

Tsinghua University

Beijing, China

M.S. Environmental Sciences and Engineering; outstanding graduate

June 2020

Nanjing University

Nanjing, China

B.S. Environmental Engineering; outstanding graduate

July 2017

PUBLICATIONS

Machine Learning for Climate Science

Liu, S., Paul A. O'Gorman. CERA: A Framework for Improved Generalization of Machine Learning Models to Changed Climates. *Submitted to Journal of Advances in Modeling Earth Systems*.

High Resolution Climate Simulations

Liu, S., Zeman, C., & Schär, C. (2025). Assessing cloud feedbacks over the Atlantic with bias-corrected downscaling. *Journal of Advances in Modeling Earth Systems*, 17, e2024MS004661.

Liu, S., Zeman, C., & Schär, C. (2024). Dynamical Downscaling of Climate Simulations in the Tropics. *Geophysical Research Letters*, 51, e2023GL105733.

Liu, S., Zeman, C., Sørland, S. L., & Schär, C. (2022). Systematic Calibration of a Convection-Resolving Model: Application over Tropical Atlantic. *Journal of Geophysical Research: Atmospheres*, e2022JD037303.

Air Pollution and Human Health

Liu, S., Xing, J., Wang, S., Ding, D., Cui, Y., & Hao, J. (2021). Health benefits of emission reduction under 1.5°C pathways far outweigh climate-related variations in China. *Environmental Science & Technology*, 55(16), 10957-10966.

- Liu, S.,** Xing, J., Wang, S., Ding, D., Chen, L., Hao, J. (2020). Revealing the impacts of transboundary pollution on PM_{2.5}-related deaths in China. *Environment International*, 134, 105323.
- Xing, J., Lu, X., Wang, S., Wang, T., Ding, D., Yu, S., ... & Hao, J. (2020). The quest for improved air quality may push China to continue its CO₂ reduction beyond the Paris Commitment. *Proceedings of the National Academy of Sciences*, 117(47), 29535-29542.
- Wang, S., **Liu, S.** (2019). Air pollution and lung cancer risks. In: Nriagu, J.O. (Ed.), *Encyclopaedia of environmental health, 2nd Edition*. Elsevier Science.
- Wang, Y., Liu, Y., **Liu, S.,** Wang, B., Duan, X., Wang, S., & Zhao, B. (2025). Changing climate alters the time people spend outdoors. *Sustainable Horizons*, 13, 100133.
- Yao, M., Niu, Y., **Liu, S.,** Liu, Y., Kan, H., Wang, S., ... & Zhao, B. (2023). Mortality burden of cardiovascular disease attributable to ozone in China: 2019 vs 2050. *Environmental Science & Technology*, 57(30), 10985-10997.
- Jiang, Y., Ding, D., Dong, Z., **Liu, S.,** Chang, X., Zheng, H., ... & Wang, S. (2023). Extreme emission reduction requirements for china to achieve world health organization global air quality guidelines. *Environmental Science & Technology*, 57(11), 4424-4433.
- Liu, Y., **Liu, S.,** Wang, S., & Zhao, B. (2022). How will window opening change under global warming: A study for China residence. *Building and Environment*, 209, 108672.
- Liu, S., Xing, J., Sahu, S. K., Liu, X., **Liu, S.,** Jiang, Y., ... & Wang, S. (2021). Wind-blown dust and its impacts on particulate matter pollution in Northern China: current and future scenarios. *Environmental Research Letters*, 16(11), 114041.

CONFERENCE PRESENTATIONS

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| American Geophysical Union Meeting | USA |
| <i>Oral presentation</i> | Dec 2023 |
| <ul style="list-style-type: none"> Bias-corrected Dynamical Downscaling: Motivation, Methodology and Application to Cloud Feedbacks in the Tropics | |
| European Geophysical Union Meeting | Austria |
| <i>Oral presentation</i> | April 2023 |
| <ul style="list-style-type: none"> Understanding the Double-ITCZ Problem over the Atlantic with Bias-corrected Downscaling | |
| European Geophysical Union Meeting | Austria |
| <i>Oral presentation</i> | May 2022 |
| <ul style="list-style-type: none"> Systematic Calibration of A Convection-Resolving Model: Application over Tropical Atlantic | |

TEACHING EXPERIENCES

Bachelor's Courses

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| Risk analysis of environmental health | Tsinghua University |
| Numerical method in environmental physics, Environmental Systems, Python in Geosciences | ETH Zürich |

Master's Courses

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| Numerical modeling of weather and climate models | ETH Zürich |
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Bachelor's Thesis

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| Visualisation of simulated clouds over the Atlantic | ETH Zürich |
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Master's Thesis

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| Extreme Heat Waves Over the Persian Gulf | ETH Zürich |
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SERVICES

Reviwer for <i>Earth's Future</i>	Since 2025
Organizer of monthly <i>Machine Learning and Climate Meetups</i> in the Greater Boston area, engaging participants from MIT, Harvard, Google, Boston University, and Boston College across all career stages.	Since 2025